The rule base is the program

formal guardrails for LLM agents, act II

Three services — a CMS, a ticket desk, a receivables tracker — written as rules, served by one domain-free engine, with a solver reviewing every change.

Research review with code: the developer experience, two sabotage episodes, one domain transfer — and act I (proofs beside the code) in a single slide. August 2026.

No formal-methods background assumed — the four concepts you need are on slide 3.

WHY THIS PROJECT

Agents now out-write our ability to check

- LLM agents produce code faster than humans can meaningfully review it. Human attention is the bottleneck — and it samples, it does not cover.
- Tests also sample: they check the runs you thought of. The bugs that matter live in the interactions you did not think of.
- If we want agents (and background jobs, and time) to act autonomously, we need a way to say what must never break — and have a machine enforce it.

The proposal

The developer writes down intent as rules and invariants. Agents write the rest. Reasoning engines sit between them and arbitrate: every change must provably keep the rules.

Code becomes cheap to regenerate. The spec becomes the durable asset.

Setting: a regular web SaaS application — CRUD/API handlers, authorization, tenancy, workflows, background jobs. Not kernels, crypto, or avionics: every technique here is judged against ordinary product teams, CI budgets, and mainstream languages at the edges.

BACKGROUND IN FOUR WORDS

All the formal methods you need today

Invariant

A sentence about the system that must be true at every moment. Example: “a user never sees another user's data.” You already write these — in comments, runbooks and post-mortems. Here they become machine-checkable.

Model

A small, faithful board-game version of your system: its states and legal moves. In act II the “model” is not beside the system — the rule base IS both the model and the running program.

Checker

A tireless adversary (here: the Z3 solver). It considers every situation the rules allow — thousands you would never test — hunting for one that breaks an invariant.

Counterexample

The checker's proof of failure: the exact situation that breaks the rule, with the rule that allowed it named. Not “something is wrong” but “an editor who authored this article can publish it, via editor_decide.”

ACT I IN ONE SLIDE (2026-07 → 08-06)

Proofs beside the code: it works — and it has a tax

2 protocols falsified, then proven

P1: the model checker beat a careful engineer twice on an online DB migration (drain guard, IS-NULL backfill). Version 3 survived — and was later proven inductively, for any system size, and live (tracks I, J, M).

59 anomalies per run

P3: skip the guard against real Postgres under concurrent load and the model's predicted anomaly appears 59×/run; 0 with the proven choreography. The invariant is load-bearing, not decorative.

1 round to the full fix

P4: same bug, same model, same generic prompt — a stronger frozen gate turned a partial fix into the full fix. Invest in gate strength, not prompt steering.

3.94× faster, never wrong

P5: an agent optimized the CMS freely inside the frozen gate; two correctness-breaking “optimizations” were absorbed mechanically. Counterexamples, not humans, did the reviewing.

Proofs at five levels

Inductive invariants, parameterized (machine-inferred by UPDR), liveness under fairness, noninterference via self-composition, and a Dafny proof of real session code (tracks I–M).

The residual tax

Every result above polices the model↔code boundary: MBT, trace validation, extraction fidelity. Act II removes that boundary for the ruled part of the system.

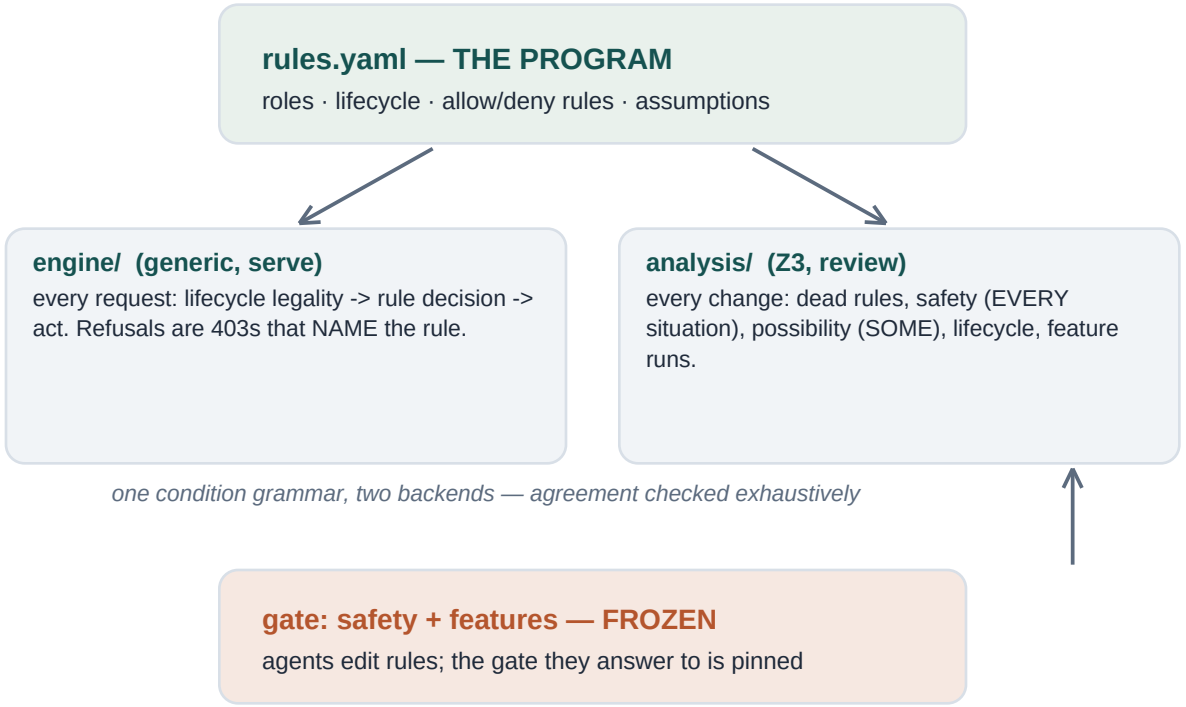
Language scouting closed the loop (notes 11–12): F* is not the SaaS rule language; of five ways proofs can meet code, “verified engine + small analyzable DSL” — Cedar's shape — ranked first. Act II builds exactly that.

ACT II · THE REFRAMING

“Is formal methods even the right framing?”

The developer's question, verbatim: all the examples model one facet of a system. If a human builds a full web service from the ground up — maybe formal methods is not the right framing. How about rule-based systems?

- Rule engines (OPS5, CLIPS, Drools, DMN) ran real businesses — and died of interaction opacity: at 10^3 rules nobody could ask “does any rule ever let X happen?”. That question is exactly an SMT workload.
- Formal methods stall on the opposite problem: the spec and the code are two artifacts, and they drift. Executable rules remove the seam — there is only one artifact for the domain.



The synthesis (note 13): rules are the programming surface · the solver is the reviewer · proof effort concentrates on the once-proven engine (Cedar's shape, note 12 pattern 1). Time, relations and computation still escalate up the assurance ladder.

A feature is a rule, not a handler

The whole CMS is one reviewable file: lifecycle plus fourteen allow/deny rules, each carrying the stakeholder sentence it translates. There are no handlers to write — endpoints exist because the lifecycle says so.

```
# rulesets/cms/rules.yaml — the CMS (130 lines)
lifecycle:
  transitions:
    - {action: submit,  from: draft,     to: in_review}
    - {action: publish, from: in_review, to: published}
    - {action: archive, from: published, to: archived}

- id: no_self_decision
  description: "Nobody may publish or reject an
    article they authored themselves — separation
    of duties, with no exception for admins."
  effect: deny
  when: 'action in ["publish", "reject"]
         and actor.is_author'
```

```
$ curl -sX POST $CMS/articles/7/publish -H 'X-User: ed'
HTTP/1.1 403 Forbidden
{
  "error": "forbidden",
  "denied_by": "no_self_decision",
  "description": "Nobody may publish or reject an
                  article they authored themselves…",
  "situation": {"role": "editor", "action": "publish",
                "is_author": true, "state": "in_review"}
}
```

- One vocabulary end to end: the ticket sentence, the rule id, the solver's finding, and the 403 body all say no_self_decision.
- Deny beats allow, silence means deny (Cedar/XACML semantics, stated once) — so admins are denied too: policy, not privilege, decides.

DEVELOPER EXPERIENCE · THE GATE

What must never happen — and what must keep working

The gate is two YAML files, frozen for agents. Safety quantifies over EVERY situation the rules allow; possibility demands witnesses (a “fix” that makes the system safely do nothing fails); features are frozen scenarios whose refusals are expected by name.

```
# safety.yaml — FROZEN for agents editing rules
safety:
  - id: S2_separation_of_duties
    requires: 'implies(action == "publish",
                       not actor.is_author)'
possibility:            # the anti-“does nothing” half
  - id: P2_review_by_non_author
    witness: 'action == "read" and not actor.is_author
              and resource.state == "in_review"'
lifecycle:
  only_into: {published: [publish]}

# features.yaml — refusals expected BY NAME
- {actor: ed,   action: publish,
   expect: deny,  denied_by: no_self_decision}
- {actor: erin, action: publish,
   expect: allow, state_after: published}
```

```
$ python -m analysis.analyze rulesets/cms
rules: 14 (5 deny, 9 allow) | situations: 8640
-- dead rules --
   ok: all 14 rules are effectual
-- assumptions --
   ok: creating roles are all assumable authors
-- safety: must hold in EVERY allowed situation --
   ok  S1 … S9   (9 properties)
-- possibility: must hold in SOME situation --
   ok  P2_review_by_non_author
       witness: {role: editor, action: read, …}
-- lifecycle --
   ok: 6 transitions live, gated entries respected
-- feature runs (frozen scenarios) --
   ok  5 features (34 steps)
VERDICT: PASS (0 findings)
```

- This runs in CI on every rule change, in seconds. Merging a rule diff means: no dead rules, every safety property proven over all situations, every workflow still alive, every scenario still passing.

Two plausible edits, seven named findings

Sabotage episode 1: two locally-reasonable edits, held to the frozen gate (--gate points at the pinned copy).

```
# edit 1: legal adds a privacy rule (LEG-77)          # edit 2: separation of duties removed
+ - id: strict_privacy
+   effect: deny
+   when: 'action == "read" and resource.state != "published" and not actor.is_author'
- - id: no_self_decision                               # “admins found it annoying”
```

```
$ python -m analysis.analyze rulesets/cms-buggy --gate rulesets/cms
FAIL DEAD allow rule 'editor_read_all': it never grants anything
     (overridden where it overlaps: deny_inactive, strict_privacy)
FAIL S2_separation_of_duties
     counterexample: {role: editor, action: publish, is_author: true, state: in_review, …}
     allowed by: editor_decide
FAIL P2_review_by_non_author: IMPOSSIBLE — blocked by deny rule(s): strict_privacy
FAIL feat_publish_lifecycle: step 4 (ed read): expected allow, got deny (rule: strict_privacy)
VERDICT: FAIL (7 findings)
```

- Four different detectors fire at once: the dead-rule check catches the silent masking that rotted 1980s rule bases; ∀-safety catches the removed deny with a concrete situation and the granting rule named; ∃-possibility catches the workflow the new deny killed — the classic safety-only blind spot; and the frozen features catch both, by name.

EPISODE · BACKGROUND PROCESSING

A background job is just another actor

Ticket SYND-9: import published articles from publisher feeds, nightly. The importer gets no back door — it is an unprivileged HTTP client (role importer), so everything interesting about it is policy:

```
# the pipeline cannot skip provenance…
- id: import_needs_provenance
  effect: deny
  when: 'actor.role == "importer"
         and action == "create"
         and not resource.has_source'

# …and a stolen importer credential is contained:
- id: importer_scope
  effect: deny
  when: 'actor.role == "importer"
         and action not in ["create", "read",
                            "submit"]'
```

```
$ analyze rulesets/cms-import-naive --gate rulesets/cms
  # the obvious draft: syndicate straight to published
  # (“the publisher already reviewed it”)
FAIL role 'importer' can create items, but the
     assumptions say it can never be an author —
     stale assumption: analysis would silently
     skip every importer-authored situation
FAIL lifecycle: 'syndicate' (draft -> published)
     enters 'published'; gate allows only ['publish']
FAIL S8_imports_have_provenance  (+S9, +feature)
VERDICT: FAIL (5 findings)
```

- Falsifier RB1 (tickets land as rule diffs): held — engine/ untouched; the change was 1 role, 1 field, 3 rules, 2 clients.
- The episode forced two new generic gate kinds: stale-assumption detection (the silent-unsoundness class) and gated lifecycle entries — review cannot be skirted by inventing a new transition into published.

EPISODE · DOMAIN TRANSFER

Same engine, different world: money you are owed

The transferability test, from the ticket: “the user describes the amount they're owed, the approximate payer name or exact payment reference, and the due date; the system provides a dashboard and email notifications about overdue payments; bank transaction emails settle claims.”

```
projections:            # time enters the vocabulary
  - {name: is_past_due, kind: date_passed,
     field: due_date}

# the ticket sentence, as a rule:
- id: claim_needs_identification
  effect: deny
  when: 'action == "create"
         and not (resource.has_payer_name
                  or resource.has_reference)'

# even the clock bot obeys the calendar:
- id: no_premature_overdue
  effect: deny
  when: 'action == "mark_overdue"
         and not resource.is_past_due'
```

290 lines of YAML — the domain

Money truth only from the bank feed (admins bounce off only_feed_settles), absolute tenant isolation, append-only ledger, calendar law: 13 rules, 10 safety + 7 possibility properties, all solver-checked.

~100 generic engine lines — the transfer cost

One missing concept: TIME. Declared date projections + an engine clock (--today, /__clock test seam), multi-source transitions (settle from awaiting AND overdue), a feature-file clock (advance_days).

Client-side, correctly — the projection boundary

Email parsing, the fuzzy matching itself (exact reference, else normalized payer name + exact amount), reminder dedup, cron. Cross-item and approximate: beyond any per-situation vocabulary.

- The transfer worked and its cost is measurable — and conceptual, not volumetric: the engine was missing time, not receivables.

The demo transcript, verbatim

```
$ python receivables_demo.py                              (engine clock: 2026-08-11)
-- users register what they are owed --
   ok  rita's dashboard shows exactly her 3 claims (uma's is invisible)
-- day 1: the bank's transaction emails arrive --
   ok  matched by approximate name (JOHN SMITH) and exact reference (INV-2026-001)
   ok  the stranger's transfer is held for manual review, settles nothing
-- day 1: the overdue sweeper runs (it attempts EVERY awaiting claim) --
   ok  every attempt refused by name: [(3, 'no_premature_overdue'), (4, …)]
-- 30 days pass (clock -> 2026-09-10); the sweeper runs again --
   ok  claim 3 (due 08-15) is overdue; uma's (due 09-30) is protected
         reminder -> rita@example.com: Payment overdue: EUR 500.00 (due 2026-08-15)
-- day 2: the late payment finally lands --
   ok  ACME S.R.O. matches the overdue claim 3; settle succeeds
   ok  even the admin cannot fake money truth: 403 only_feed_settles
VERDICT: PASS
```

- The sweeper is deliberately dumb — it tries everything, and the rules hold the clock accountable, not vice versa. A buggy (or compromised) bot is bounded by policy the solver already checked over every situation.
- Late payments land because possibility W2 (“settle from overdue exists”) is part of the frozen gate — liveness guarded the design before the demo existed.

SCOREBOARD

Three services, one engine, every claim executable

3 : 1

services per engine — CMS, tickets, receivables on 1,097 lines of domain-free Python (grep-provable: zero domain words outside docstrings).

313 : 1,097

lines of YAML that ARE the CMS vs the shared generic engine+analyzer. Receivables: 290. Tickets: 100.

36,096

situations checked exhaustively — runtime evaluation and the Z3 compilation agree on every one (8,640 + 576 + 26,880).

12

named findings across two sabotaged rule bases (7 + 5); zero false passes; every finding carries a rule name or a concrete situation.

What the analyzer asks of every rule-base change (seconds, in CI):

- Is any rule dead — does removing it change no decision? (rule rot, reversed: two redundant guards were proven useless and deleted)
- Do the safety properties hold in EVERY allowed situation? If not: the situation + the granting rule.
- Is every workflow still POSSIBLE? If not: the blocking deny, found by re-checking without each deny.
- Are the assumptions consistent with the rules? (every creating role must be an assumable author)
- Is the lifecycle live and are gated states entered only through their gate (only_into)?
- Do the frozen feature scenarios still run — with every expected refusal denied by the right rule?

WHERE IT STOPS

Kept deliberately sharp

The projection boundary

Rules see finite booleans (is_author, has_source, is_past_due). Fuzzy matching, cross-item invariants (“no two claims with one reference”, “≤3 published per author”) live in clients today — properly in store constraints (P9: the invariant→Postgres compiler that doesn't exist yet, anywhere).

No time-interleaving

Single-situation rules can't see races: the demoted author who is still is_author, the stale session. That is rung 2's job — the Quint temporal twin (falsifier RB4) stays on the ladder.

The clock is a trust root (RB6)

is_past_due is only as true as the engine's date. The analyzer proves “overdue only when past due” relative to the projection; a skewed clock breaks it silently. The temporal twin should model skew.

One entity per rule base

Receivables dodged it (transactions live with the bank; only claims are ruled). Linked ruled entities — invoice↔payment↔dunning — need N rule bases plus client joins, or engine work on relations.

The engine is unproven Python

Fine for research; the production form is a verified engine — Cedar's Lean proofs, or our own track-D Dafny→Go shape — so trust rests on one proof plus per-change analysis.

GUARDRAILS · CLAUDE.MD

What we learned the hard way — now enforced

1 · The spec is frozen for agents

Enforced mechanically — the analyzer takes --gate from a pinned directory; agents edit rules, never the gate. Same contract as act I's frozen invariants.

2 · Gates need both directions

Safety-only gates accept fixes that trade away liveness (“the safest system does nothing”). Every gate here = safety + possibility witnesses + frozen feature runs.

3 · Gate strength beats NL steering

Same bug, model, prompt: the stronger gate turned a partial fix into the full fix (P4). Cheaper models are fine when the gate is strong.

4 · Escalate to proofs

Rules (this deck) < temporal models < inductive < parameterized < liveness < hyperproperties. Climb only when the ticket needs time, relations, or computation.

5 · Translate / validate split

LLM translation of NL→rules is unsound — always followed by a sound solver step. Humans review rules sentence-by-sentence, never agent edits.

6 · Counterexamples are the currency

Machine-readable and named: unsat blockers, situation witnesses, denied_by 403s. One vocabulary from ticket to runtime error.

7 · Verify sub-agent claims

The checker corrected the author repeatedly (two “correct” protocols, two “needed” rules). Dead ends and falsified predictions are recorded in the notes.

8 · Boundaries by construction

Bots are ordinary actors with no back door; capability tokens make bypass a compile error (track G). Typestate over discipline.

ROADMAP

Falsifiers on the table

Held so far

- RB1 (tickets land as rule diffs, engine untouched): held twice — imports cost zero engine lines; the receivables transfer cost ~100 generic lines for one missing concept (time). Prediction: gate-language growth flattens as new domains reuse projections, only_into, the assumption check.
- Act I stands underneath: the migration protocol and CMS model remain proven at five levels (tracks I–M); the P4/P5 loop is the template for RB5.

Next

- RB3 — the LLM fidelity episode: same tickets, ticket→rule-diff vs ticket→handler-code; measure human corrections needed. The bet: declarative diffs review better.
- RB5 — agent repair: hand an agent the sabotaged rule bases under the frozen gate; expect ≤2 rounds (P4's result, cheaper, because diffs are rules).
- RB4 — the temporal twin in Quint: the demoted-author race and clock skew that per-situation rules provably cannot see.
- RB2 — scale: exhaustive backend agreement is 26,880 situations (~80 s) for receivables; per-condition projected enumeration or symbolic equivalence when it hurts. P8 — the same rules as Cedar policies (verified engine, symbolic analysis). P9 — the invariant→Postgres-constraint compiler nobody has built.

Parked (consciously, with reasons recorded)

Alloy (Z3 covers it) · Quint→Rust codegen (no tooling) · Kani until unbounded domains arrive · Verus (proxy-blocked; Dafny stands in).

The one-sentence takeaway

The developer writes the rules and the gate; the solver reviews every change;
agents, bots, and time do their worst inside the fence.
For the ruled part of the system there is no code to review — only a rule diff and a verdict.

Try it: examples/rule-driven-cms/check.sh — tests + 5 analyses + 5 live demos, one command

Read it: research/13-rule-based-cms.md (the framing + both episodes) · research/INDEX.md (all notes)

Act I evidence: research/09, 10, 12 · prototypes p1–p7 · examples/cms (the spec-beside-code twin)